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## Education

### PhD in Applied Mathematics

Nov.2021–Oct.2024 **Politecnico di Torino**, Dipartimento di Scienze Matematiche "Luigi Lagrange", *Cum laude*.

Thesis title: *Machine Learning algorithms in Computational Fluid Dynamics: Improving Reynolds-Averaged Navier-Stokes equations by ML closure models*, Supervisor: Prof. Stefano Berrone.

2023–2024 **Sauber Motorsport**, Two visiting periods in Formula 1 team in the Computational Fluid Dynamics Methodology group, Total of 6 months.

### Master's degree in Mathematical Engineering

2018–2020 **Politecnico di Torino**, GPA: 110 cum laude/110.

Thesis Title: "Computational simulation of the flow around rectangular cylinders: Effects of grid quality at wall". Thesis advisor: Prof. Luca Bruno

Sep.2019–Feb.2020 **École Centrale de Nantes**, ERASMUS semester in the "Computational Mechanics" Master's degree class.

### Bachelor's degree in Mathematical Engineering

2015–2018 **Politecnico di Torino**, GPA: 108/110.

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## Research interests

**Scientific machine learning** for problems described by Partial Differential Equations, **Model Order Reduction**, **Computational Fluid Dynamics**, any intersection of the three.

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## Academic positions

Nov.2024–Current **Research fellow**, MathLab group at Scuola Internazionale Superiore di Studi Avanzati (SISSA) at Gianluigi Rozza's group, Research topics: Scientific Machine Learning, non-intrusive Reduced Order Modeling, Computational Fluid Dynamics.

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## Publications

### Journal Articles

2025 Davide Oberto, Davide Fransos and Stefano Berrone, *Using Delayed Detached Eddy Simulation to create datasets for data-driven turbulence modelling: a periodic hills with parameterized geometry case*, Computers & Fluids.  
doi:10.1016/j.compfluid.2024.106506

2022 Stefano Berrone and Davide Oberto, *An invariances-preserving vector basis neural network for the closure of Reynolds averaged Navier-Stokes equations by the divergence of the Reynolds stress tensor*, Physics of Fluids, 34.  
doi:10.1063/5.0104605

2022 Luca Bruno and Davide Oberto, *Effects of cell quality in grid boundary layer on the simulated flow around a square cylinder*, Computers & Fluids, 238:105351.  
doi:10.1016/j.compfluid.2022.105351

### Chapters

2025 Davide Oberto, *Improving the Vector Basis Neural Network for RANS Equations Using Separate Trainings*, Numerical Mathematics and Advanced Applications ENUMATH 2023, Volume 2.

2025 Stefano Berrone, Davide Oberto, Moreno Pintore and Gioana Teora, *The lowest-order Neural Approximated Virtual Element Method*, Numerical Mathematics and Advanced Applications ENUMATH 2023, Volume 1.

### Preprints

Davide Oberto, Maria Strazzullo and Stefano Berrone, *Vector basis neural network for the closure of Parametric Reynolds averaged Navier-Stokes equations at full and reduced order levels*, <https://arxiv.org/abs/2510.06992>.

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## Teaching and supervision

### Teaching assistant

Topics in Computational Fluid Dynamics

**PhD class at SISSA**, *Total of 5 hours*, Main topics: Discretization through Finite Element Method of: Stokes equations (inf-sup condition and stabilization techniques), Navier-Stokes equations (fixed point and Newton methods to deal with nonlinearity and time discretization).

Numerical Methods and Scientific Computing

**1<sup>st</sup> year in Aerospace Engineering masters at PoliTo**, *Total of 60 hours*, Main topics: Iterative Methods for solving Linear Systems, Preconditioning, Non-linear equations, ODEs, FD for 1D and 2D problems, FEM for 1D problems, numerical schemes for 1D conservation laws.

Numerical Linear Algebra

**1<sup>st</sup> year in bachelor's programs at PoliTo**, *Total of 80 hours*, Main topics: Floating-point representation, Interpolation through polynomials and splines, Regression, Direct Methods for solving Linear Systems, Eigenvalues computation and Singular Value Decomposition.

### Supervision

PhD Currently co-supervising one PhD student on digital twins for cardiovascular problems.

MSc Currently supervising two MSc projects on CFD related topics.

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## Revisions

Revised one paper for **Journal of Computational Physics**.

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## Conferences

Attended as contributed or invited speaker at several national and international conferences.

ParCFD **33rd Parallel CFD International Conference**, Alba, 2022.

UMI Workshop **Matematica per l'Intelligenza Artificiale e il Machine Learning Giovani ricercatori**, Torino, 2022.

- ENUMATH **European Conference on Numerical Mathematics and Advanced Applications**, *Lisbon*, 2023.
- ECCOMAS **9th European Congress on Computational Methods in Applied Sciences and Engineering**, *Lisbon*, 2024.
- SIMAI **Società Italiana di Matematica Applicata e Industriale**, *Trieste*, 2025.
- YIC **Young Eccomas Investigators**, *Pescara*, 2025.

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## Personal skills

- Coding **Bash**, *Basic knowledge*, Scripting.  
**C++**, *Intermediate knowledge*, Mainly as background language for OpenFoam.  
**Python**, *Everyday user*, Scripting and AI tools development.  
**Matlab**, *Everyday user*, For numerical methods code development.
- AI **Tensorflow and PyTorch**, Developed several neural network models for my research.
- OS **Linux**, *Everyday user*, All my research performed in Ubuntu environment.  
**Windows**, *Everyday user*.
- CFD **OpenFoam**, *Everyday user*, Implemented custom turbulence models and function objects.